

Robert C. Wong

robwong15@gmail.com | linkedin.com/in/robertcwong | github.com/robw0ng | robw0ng.com

EDUCATION

Stony Brook University

Bachelor of Science in Computer Science

Stony Brook, NY

May 2026

SKILLS

Languages: TypeScript, Python, C#, C, Java, JavaScript, HTML/CSS, SQL

Frameworks: React, React Native, GraphQL, Flask, Express, Node.js, ASP.NET, EF-Core

Developer Tools: Git, Kafka, Figma, Storybook, Azure, Docker, Postman, MongoDB, Linux, VS Code, JetBrains IDEs

EXPERIENCE

Umbrava

Software Engineer

Hauppauge, NY

Aug. 2025 - Present

- Owned an inventory and asset management platform end-to-end across React web and React Native mobile, scaling to **250,000+ items per tenant** with offline-first writes, dead-letter retry, and real-time sync over GraphQL.
- Built semantic deduplication across **1M+ work orders**, running nearest-neighbor search on OpenAI text-embedding-3-small vectors in SQL Server with no external vector store, prefiltering to bound scan cost.
- Built a live scheduling and dispatch board with optimistic drag-and-drop assignment, validating trips against SLA windows and priorities defined per cross-tenant contract, across React TypeScript, C# REST APIs, and T-SQL.
- Shipped the team's first native iOS module, bridging React Native to Swift with ActivityKit to surface real-time status updates on the lock screen.

Software Engineer Intern

Jun. 2025 - Aug. 2025

- Delivered an embedded analytics module on Logi Symphony, including a dashboard list, embedded viewer, access control sharing modal, and scheduled email delivery for over **20,000** clients and vendors.
- Built a super-admin tooling suite from scratch, including multi-tenant user management, bulk vendor connections, and per-tenant configuration, each with permission-gated access and audit logging.
- Implemented Kafka event producers and handlers to publish and sync messages across microservices, ensuring data consistency between distributed databases, with retry policies, dead letter queues, and health-check-driven auto-rollback to the last passing build.

NYPD Compliance Division

Software Development Intern

Manhattan, New York

Jul. 2024 - Aug. 2024

- Processed millions of body cam interaction records monthly across **35,000+ NYPD officers** into clean, validated datasets powering automated reporting pipelines used by department chiefs.
- Delivered custom software that **improved departmental efficiency by 86%**, replacing manual workflows.
- Developed and led 2 large-scale projects, a custom data entry application for clerks and a data processing pipeline that cleaned, validated, aggregated, and presented data for department chiefs, both now in daily use.

PROJECTS

Vetty | *JavaScript, Node.js, Kotlin, GraphQL, VS Code Extension API* | [GitHub](#)

Jul. 2026 – Present

- Built and open-sourced a code review tool for VS Code, Cursor, and JetBrains, porting the full review model onto the JetBrains plugin API, targeting the rapid re-review loop AI-assisted editing creates.
- Tracked reviewed state by EOL-normalized content hash so files auto-unview when edited, and re-anchored inline comments by content so review notes survive an agent rewriting the file underneath them.
- Snapshotted reviewed content into git's object store via hash-object, diffing against those blobs to show only what changed since the last pass, with merge-base resolution matching GitHub's files-changed semantics.

DroneGurus R&D | *Python, YOLOv8, OpenCV, NumPy, mmWave Radar, UWB*

Mar. 2026 – Present

- Trained a YOLOv8n thermal damage detector on a custom **27K-image** dataset with 8× colormap augmentation, reaching **91.9% mAP@50**, served via FastAPI and demoed at a trade show.
- Prototyped through-wall human detection for search-and-rescue, fusing 60GHz and 24GHz radar pairs with cross-band ghost suppression, positioned by a 4-anchor UWB array self-calibrated by classical MDS.
- Rendered an AR overlay holding targets in a fixed world frame, fusing UWB position with gyro/accelerometer pitch and roll into a view transform that projects them to pixels through calibrated intrinsics.